BLOCK 3

# General Linear Model: Linear Regression, Correlation, and t-test

## From Data Preparation to Analysis

LAST WEEK **Prepare + clean the dataset** → THIS WEEK **Use the dataset to answer research questions**

# Today’s goal: understand the logic before the code

- Recognize what correlation, t-test, and linear regression each ask.
- Choose an analysis based on the research question and variable types.
- Run all three analyses in Python using the Message Trust course dataset.
- Read the main result in plain language and make a simple visualization.

**You are NOT expected to memorize formulas or Python syntax.**

# Course dataset: remember where the data came from

A randomized experiment with two conditions and a post-message survey (N = 180)

## 1 Participants

180 people (e.g., USC students) are recruited to participate in an online study.



## 2 Random Assignment

Each participant is randomly assigned to one of two conditions (like flipping a coin).



**Control**
(standard message)
~ 90 participants

**Personalized**
(adapted message)
~ 90 participants

## 3 View the Message

Participants see a social media post based on their assigned condition.

**Control condition (standard message)**

Campus News
2h ago



Protecting our environment is important for a healthier future.

**Personalized condition (adapted to the user)**

Campus News
2h ago



As a USC student, you can help protect our environment and make a difference in our community.

## 4 Complete the Survey

After viewing the message, participants answer a short survey.

**Message trust**
(1 = strongly disagree, 7 = strongly agree)

1 2 3 4 5 6 7

**Intention to share**
(1 = strongly disagree, 7 = strongly agree)

1 2 3 4 5 6 7

**Clicked the article?**

Yes No

**Demographics (examples)**

- Age
- Gender
- Education level
- Primary social media platform
- Daily social media use (hours)

## 5 Resulting Data

Each row in the dataset represents one participant's responses and characteristics (N = 180 rows).

| ID | Condition | Age | Trust | Share | Clicked | ... |
|---|---|---|---|---|---|---|
| 1 | Control | 22 | 4.2 | 3 | No | ... |
| 2 | Personalized | 24 | 5.8 | 5 | Yes | ... |
| 3 | Control | 21 | 3.9 | 2 | No | ... |
| 4 | Personalized | 27 | 6.1 | 6 | Yes | ... |
| 5 | Control | 19 | 2.7 | 1 | No | ... |
| ... | ... | ... | ... | ... | ... | ... |

**Today we use this SAME dataset to ask three different statistical questions.**

# The dataset

**ROW = one participant / observation**

**COLUMN = one variable**

| ID | condition | age | gender | platform | message_trust | clicked |
|---|---|---|---|---|---|---|
| 1 | Control | 24 | Woman | TikTok | 4.2 | No |
| 2 | Personalized | 31 | Man | YouTube | 5.8 | Yes |
| 3 | Control | 27 | Woman | Instagram | 3.9 | No |

**The same dataset can answer different questions depending on which columns we compare.**

# Reminder: types of variables

Variable type helps us decide which statistical approach fits

## CATEGORICAL

categories / labels

### Nominal

No natural order
condition, gender, platform

### Ordinal

Meaningful order
education level

## NUMERIC

amounts / quantities

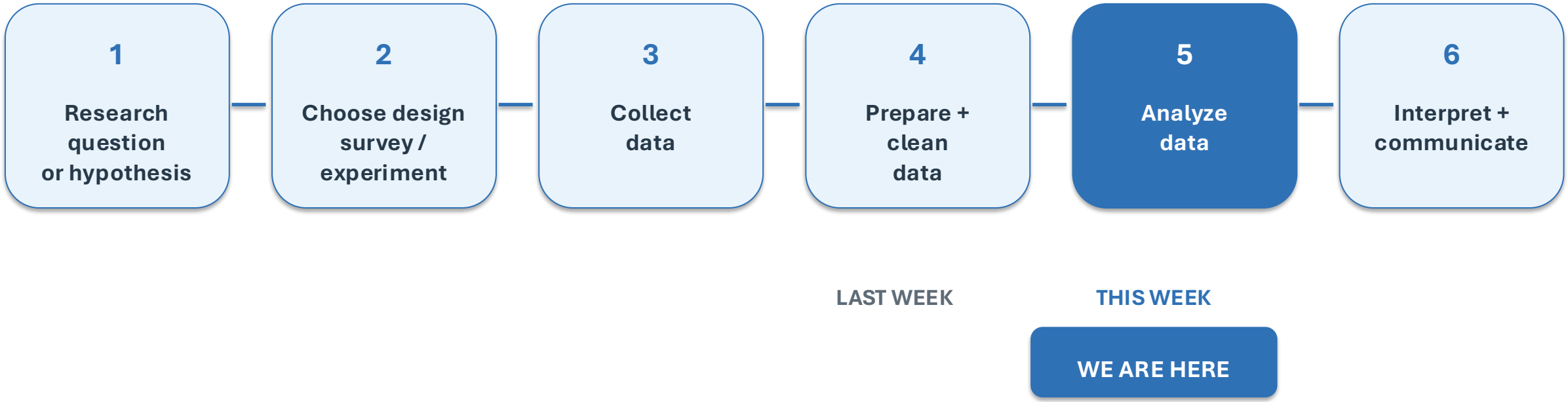
### Interval

Equal spacing
scale scores

### Ratio

Equal spacing + true zero
hours, age

# Where are we in the research process?



# What is the General Linear Model?

**The simple idea:** **Look for a pattern that helps explain why people have different scores on an outcome.**

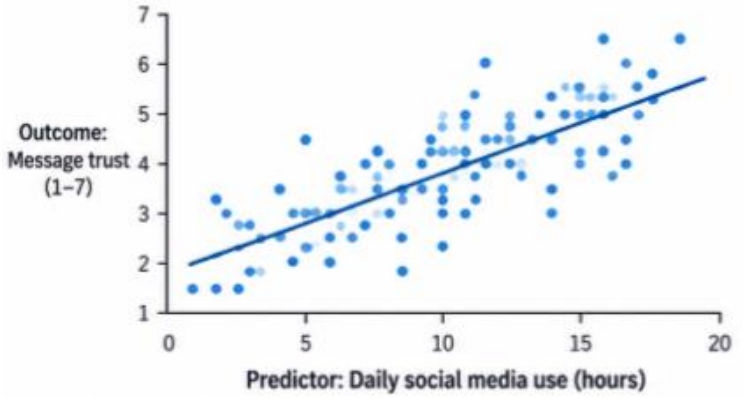
**WHAT WE OBSERVE = PATTERN + WHAT IS LEFT OVER**

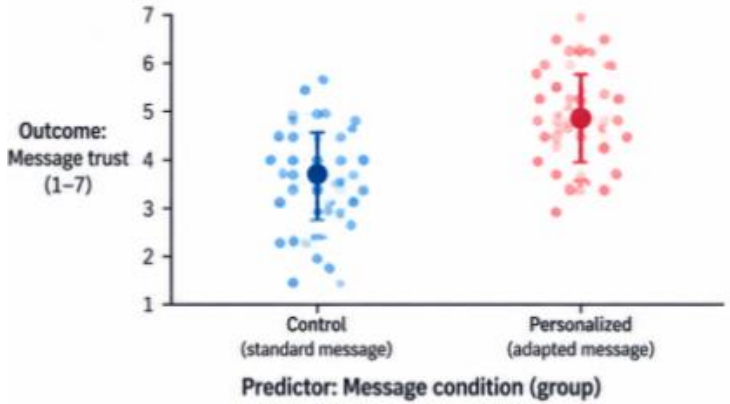
## For this class, we will connect three types of GLM:

**Correlation** — Do two numeric variables move together?

**t-test** — Do two groups have different averages?

**Regression** — How does an outcome change with a predictor?



# Three ways to ask questions with the SAME dataset

## 1 CORRELATION

**Are they related?**

baseline_trust ↔ message_trust

## 2 t-TEST

**Are the two group means different?**

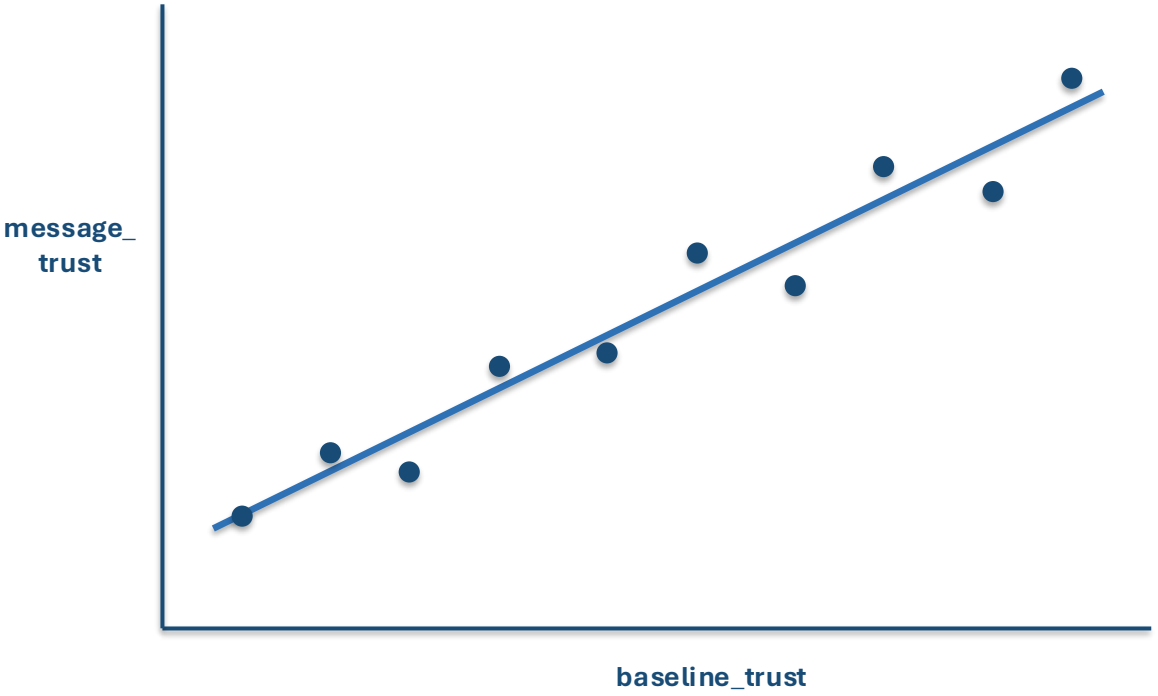
condition → message_trust

## 3 REGRESSION

**How much does Y change as X changes?**

baseline_trust → message_trust

# 1. Correlation: do two numeric variables move together?




Each dot = one participant

## Are baseline_trust and message_trust related?

- Both variables are numeric.
- r > 0: they tend to increase together.
- r < 0: one tends to decrease as the other increases.
- r ≈ 0: little linear relationship.

**Correlation gives direction + strength.**

# Open Colab: load the SAME course dataset

```
import pandas as pd

url = "https://raw.githubusercontent.com/KayChansiri/Course_research-methods-in-the-age-of-
ai/main/data/JOUR701_Block3_Course_Dataset.csv"
df = pd.read_csv(url)

df.head()
```

# Run the correlation

```
from scipy.stats import pearsonr

corr_data = df[["baseline_trust", "message_trust"]].dropna()
r, p = pearsonr(corr_data["baseline_trust"],
corr_data["message_trust"])

print("r =", round(r, 3))
print("p =", round(p, 4))
```

## What Python is doing

- Keeps the two variables we need.
- Drops rows missing either value.
- Calculates Pearson's r.
- Also gives a p-value.

**Run the cell.**
**Then read only r and p.**

# How to read a correlation result

Your notebook will print something like:

**r = +0.583  p = 0.002**

Illustrative numbers only

**1. SIGN**

\+ means the variables tend to increase together.

**2. SIZE**

|r| closer to 1 = stronger linear relationship.

**3. p-VALUE**

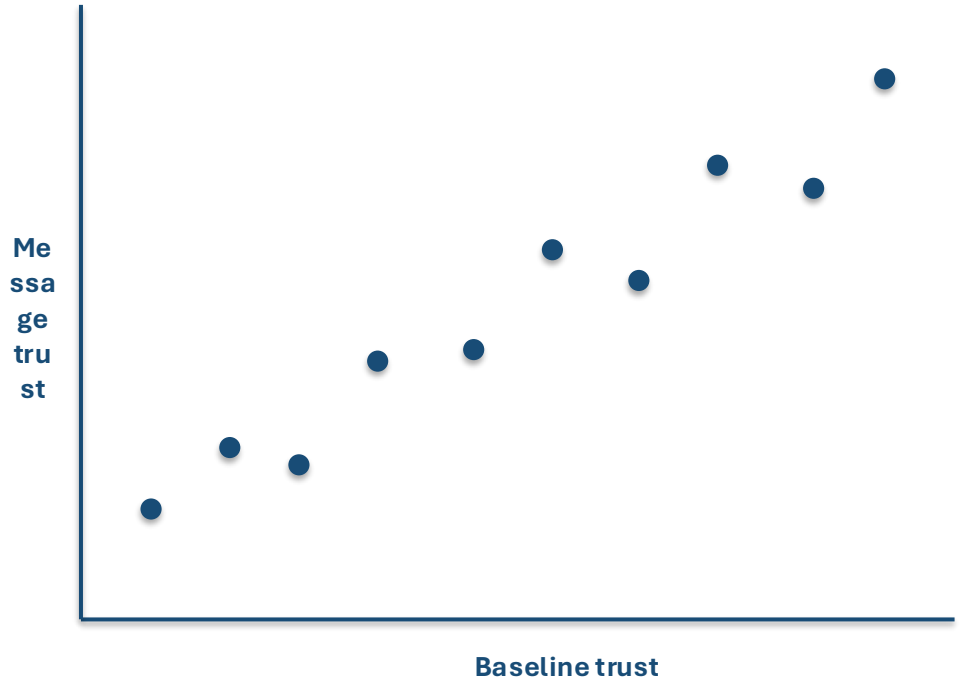
p < .05 = evidence that the correlation is not 0 in the population.

**Example interpretation: Higher baseline trust is associated with higher message trust. Correlation does NOT by itself show causation.**

# Visualize the correlation: make a scatterplot

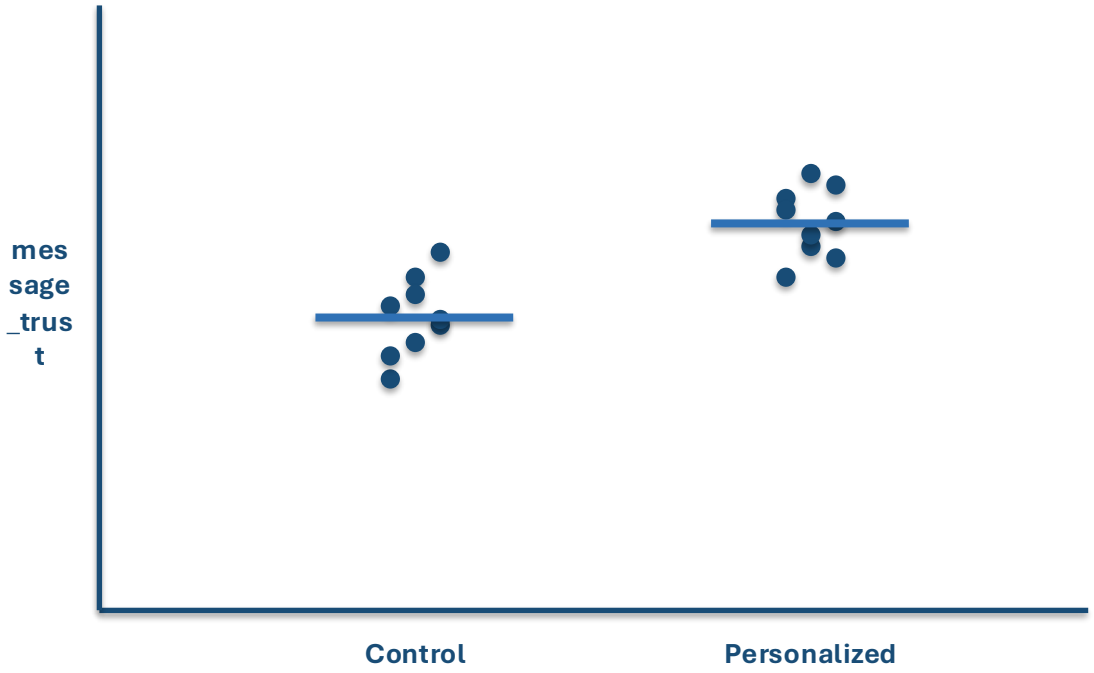
```
import matplotlib.pyplot as plt

plt.scatter(df["baseline_trust"],
df["message_trust"])
plt.xlabel("Baseline trust")
plt.ylabel("Message trust")
plt.title("Baseline Trust and Message Trust")
plt.show()
```




**Look for: direction • roughly straight-line pattern • unusual points. The plot helps you SEE what r summarizes.**

# 2. t-test: are TWO group means different?




Each dot = one participant • blue bar = group mean

## Does average message_trust differ between Control and Personalized?

- Grouping variable: condition (2 categories)
- Outcome: message_trust (numeric)
- The t-test compares the two group averages.

**Think: TWO groups → compare their means.**

# Run the t-test

```
from scipy.stats import ttest_ind

control = df.loc[df["condition"] == "Control", "message_trust"].dropna()
personalized = df.loc[df["condition"] == "Personalized",
"message_trust"].dropna()

print("Control mean:", round(control.mean(), 2))
print("Personalized mean:", round(personalized.mean(), 2))

t, p = ttest_ind(control, personalized, equal_var=False)
print("t =", round(t, 3))
print("p =", round(p, 4))
```

## Read these first

- Mean for Control
- Mean for Personalized
- p-value

The t statistic is useful, but for this class the group means + p-value are easier to interpret first.

# How to read a t-test result

**Control mean = 3.76  Personalized mean = 4.52  p = 0.018**

Illustrative numbers only

1 **Compare the means** — Which group has the higher average?

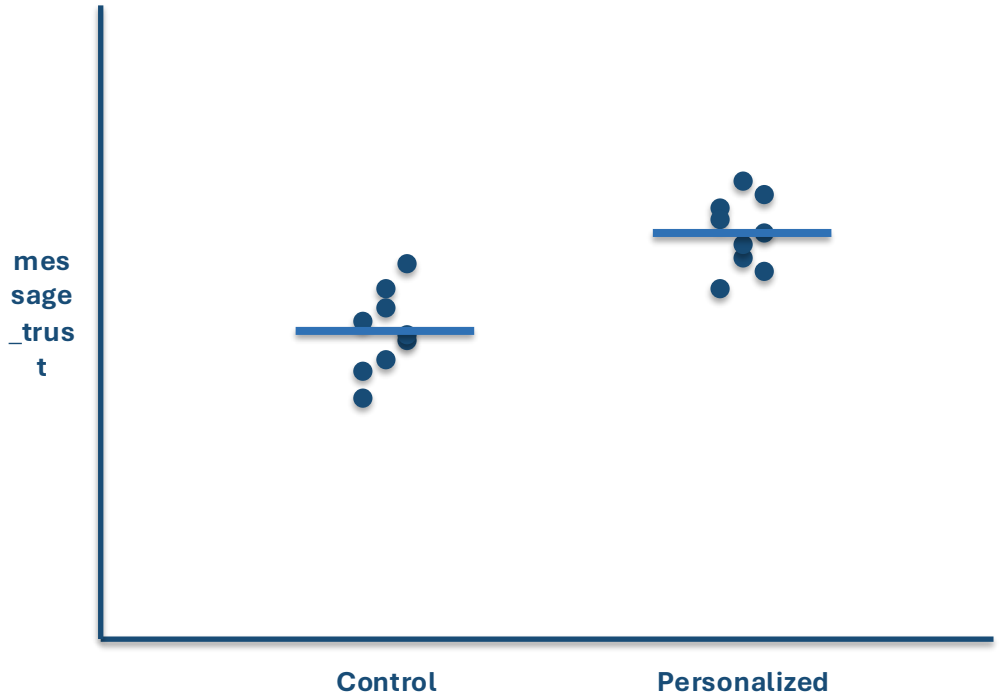
2 **Read p** — If $p < .05$, the observed difference is statistically distinguishable from 0.

3 **Write the result** — Describe which group is higher/lower and whether the evidence supports a difference.

**A significant t-test tells you the groups differ; it does not tell you the difference is large or important.**

# Visualize the t-test: compare the two groups

```
df.boxplot(column="message_trust",
by="condition")
plt.xlabel("Condition")
plt.ylabel("Message trust")
plt.title("Message Trust by Condition")
plt.suptitle("")
plt.show()
```




**The t-test gives the statistical comparison. The plot shows the distributions, overlap, and group difference.**

# 3. Linear regression: give X and Y different jobs

**Research question:** **Does baseline trust predict message trust?**

**X = predictor**

**baseline_trust**

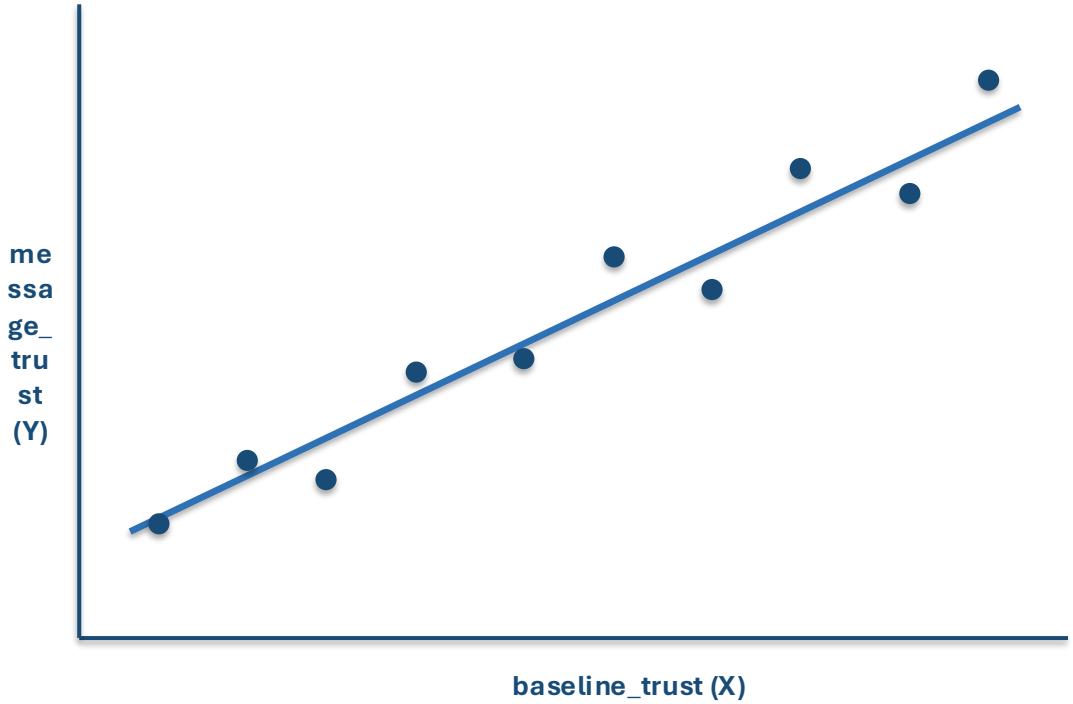


**Y = outcome**

**message_trust**

**Regression asks: how much does the expected outcome change when the predictor changes?**

# The regression line is a prediction rule




## Predicted Y = intercept + slope × X

### SLOPE / COEFFICIENT

Expected change in Y for a 1-unit increase in X

### INTERCEPT

Predicted Y when X = 0

**Focus on the slope first: direction → size → uncertainty → meaning.**

# Run the simple linear regression

```
import statsmodels.formula.api as smf

model = smf.ols(
    "message_trust ~ baseline_trust",
    data=df
).fit()

model.summary()
```

## Read the formula like English

**message_trust**
modeled using
**baseline_trust**

- Left of ~ = outcome (Y)
- Right of ~ = predictor (X)
- .fit() estimates the line

# Do NOT try to read every number in the regression table

| | |
|---|---|
| coef | How big? Which direction? |
| P>\|t\| | Is the coefficient distinguishable from 0? |
| [0.025, 0.975] | What is a plausible range for the coefficient? |
| R-squared | How much variation does the model explain? |

**For this class: coefficient → p-value / CI → $R^2$ → substantive interpretation.**

# How to read ONE regression coefficient

Illustrative output format — your Colab will produce the actual values:

**baseline_trust  coef = 0.654  p = .000  95% CI [0.519, 0.789]**

- Direction: positive → higher baseline trust is associated with higher message trust.
- Size: +1 point in baseline trust corresponds to an estimated +0.654 points in message trust.
- Uncertainty: the 95% CI does not include 0 in this illustration.
- p-value: .002 is below .05 → evidence against a zero slope.

**Do not write “X causes Y” unless the research design supports that causal claim.**

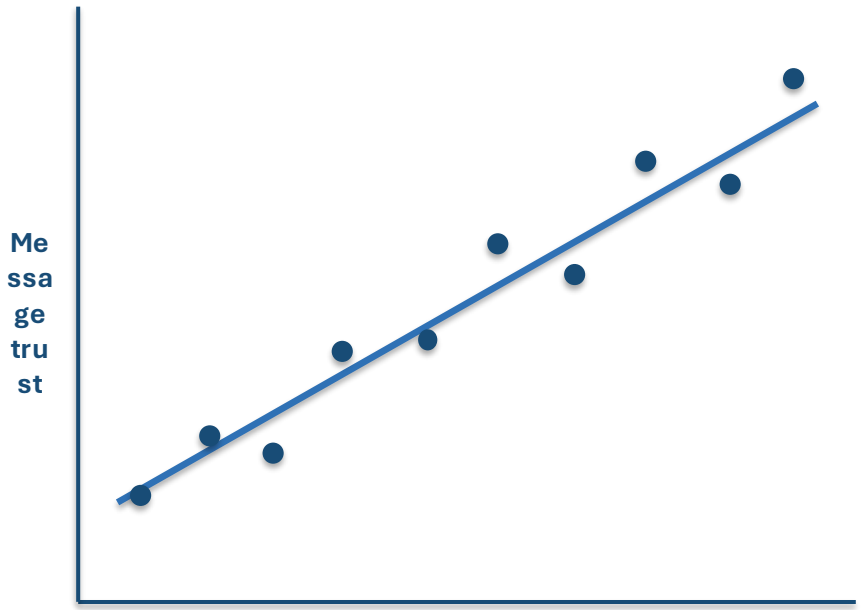
# Visualize the regression: dots + best-fitting line

```
import numpy as np
import matplotlib.pyplot as plt

plot_df = df[["baseline_trust", "message_trust"]].dropna()
plt.scatter(plot_df["baseline_trust"], plot_df["message_trust"])

m, b = np.polyfit(plot_df["baseline_trust"],
plot_df["message_trust"], 1)
x = np.linspace(plot_df["baseline_trust"].min(),
                plot_df["baseline_trust"].max(), 100)
plt.plot(x, m*x + b)

plt.xlabel("Baseline trust")
plt.ylabel("Message trust")
plt.show()
```




**Dots = observed participants**
**Line = model's predicted pattern**

# What does $R^2$ mean?

## Example: $R^2 = .339$

The model accounts for about 34% of the observed variation in message trust in this sample.

## What NOT to say

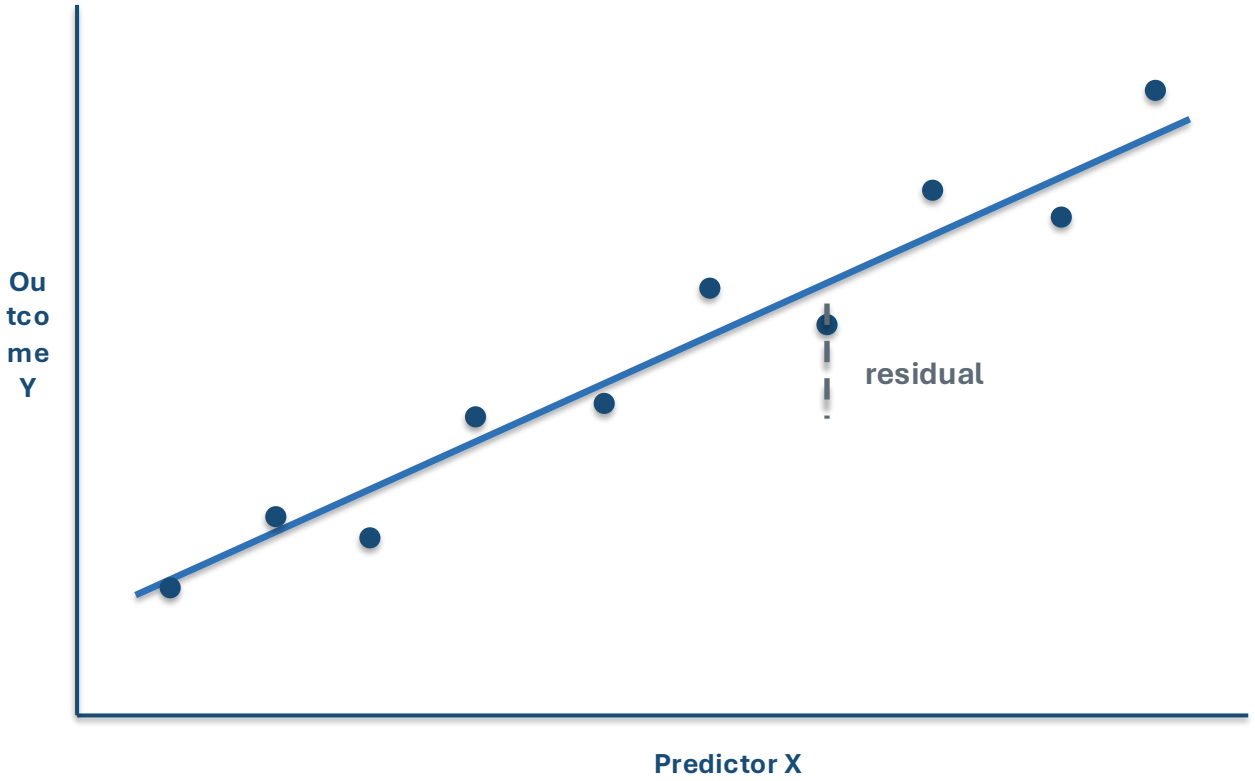
“The model is 34% accurate.”

“Baseline trust causes 34% of message trust.”

“34% of participants were predicted incorrectly.”

**$R^2$ is about variation explained by the model — not accuracy in the everyday sense.**

# What does “best-fitting line” mean?




## Each person has:

- an observed value (their dot)
- a predicted value (the line)
- a residual = observed − predicted

**The best-fitting line is the one that makes the squared residuals as small as possible overall.**

# Now the important question: which analysis should I choose?

Do NOT start by asking:

**“Which test do I remember?”**

Start by asking:

**“What is my research question, and what kinds of variables are involved?”**

**QUESTION → VARIABLE TYPES → ANALYSIS**

# A simple decision rule for today

| | | |
|---|---|---|
| **Two numeric variables** | Are baseline_trust and message_trust related? | **Correlation** |
| **Two groups + numeric outcome** | Does message_trust differ by condition? | **t-test** |
| **Predictor → numeric outcome** | Does baseline_trust predict message_trust? | **Linear regression** |

**Step 1: write the question in plain language. Step 2: identify the variable types. Step 3: choose the tool.**

# Same dataset, three different questions

| | |
|---|---|
| Are baseline_trust and message_trust associated? | CORRELATION |
| Does message_trust differ between Control and Personalized? | t-TEST |
| How much does message_trust change as baseline_trust changes? | REGRESSION |

**Different question wording + variable roles → different analytical tool.**

# Day 1 checkpoint

- When would you use a correlation?
- When would you use a t-test?
- In regression, what is X? What is Y?
- What does a positive coefficient mean?
- What does $R^2$ tell you?
- Why should you visualize the data as well as run the test?

**If you can answer these in plain language, you understand the core logic.**